

ETC5521: Diving Deeply into Data Exploration

Exploring bivariate dependencies

Professor Di Cook

Department of Econometrics and Business Statistics

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Outline

- The humble but powerful scatterplot
- Additions and variations
- Transformations to linearity
- (Robust) numerical measures of association
- Simpson's paradox
- Making null samples to test for association
- Imputing missings
- Multiple associations, exploring network data

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The world is full of obvious things which nobody by any chance observes

Quote from Arthur Conan Doyle, The Hound of the Baskervilles

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The scatterplot

Scatterplots are the natural plot to make to explore **association** between two **continuous** (quantitative or numeric) variables.

They are not just for **linear** relationships but are useful for examining **nonlinear** patterns, **clustering** and **outliers**.

We also can think about scatterplots in terms of statistical distributions: if a histogram shows a marginal distribution, a **scatterplot** allows us to examine the **bivariate distribution** of a sample.

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History

- Descartes provided the Cartesian coordinate system in the 17th century, with perpendicular lines indicating two axes.
- It wasn't until **1832** that the scatterplot appeared, when **John Frederick Herschel** plotted position and time of double stars.
- This is 200 years after the Cartesian coordinate system, and **50 years after bar charts and line charts** appeared, used in the work of William Playfair to examine economic data.
- Kopf argues that *The scatter plot, by contrast, proved more useful for scientists*, but it clearly is useful for **economics** today.

<http://www.datavis.ca/milestones/>

Features of a pair of continuous variables (1/3)

Feature	Example	Description
positive trend		Low value corresponds to low value, and high to high.
negative trend		Low value corresponds to high value, and high to low.
no trend		No relationship
strong		Very little variation around the trend

Language and terminology

Are the words “correlation” and “association” interchangeable?

In the broadest sense **correlation** is any statistical association, though it commonly refers to the degree to which a pair of variables are **linearly** related.

Wikipedia

If the relationship is not linear, call it **association**, and avoid correlated.

Feature	Example	Description
moderate		Variation around the trend is almost as much as the trend
weak		A lot of variation making it hard to see any trend

Features of a pair of continuous variables (2/3)

Feature	Example	Description
linear form		The shape is linear
nonlinear form		The shape is more of a curve
nonlinear form		The shape is more of a curve
outliers		There are one or more points that do not fit the pattern on the others

Features of a pair of continuous variables (3/3)

Feature	Example	Description
barrier		There is combination of the variables which appears impossible
I-shape		When one variable changes the other is approximately constant
discreteness		Relationship between two variables is different from the overall, and observations are in a striped pattern
heteroskedastic		Variation is different in different areas, maybe depends on value of x variable

Feature	Example	Description
clusters		The observations group into multiple clumps
gaps		There is a gap, or gaps, but its not clumped

Feature	Example	Description
weighted		If observations have an associated weight, reflect in scatterplot, e.g. bubble chart

Additional considerations (Unwin, 2015):

- **causation**: one variable has a direct influence on the other variable, in some way. For example, people who are taller tend to weigh more. The dependent variable is conventionally on the y axis. *It's not generally possible to tell from the plot that the relationship is causal, which typically needs to be argued from other sources of information.*
- **association**: variables may be related to one another, but through a different variable, eg ice cream sales are positively correlated with beach drownings, is most likely a temperature relationship.
- **conditional relationships**: the relationship between variables is conditionally dependent on another, such as income against age likely has a different relationship depending on retired or not.

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Famous scatterplot examples

Anscombe's quartet

All four sets of Anscombe has **same means, standard deviations and correlations**, $\bar{x} = 9$, $\bar{y} = 7.5$, $s_x = 3.3$, $s_y = 2$, $r = 0.82$.

Numerical statistics are the same, for very different association.

Datasaurus dozen

And similarly all 13 sets of the datasaurus dozen have **same means, standard deviations and correlations**, $\bar{x} = 54$, $\bar{y} = 48$, $s_x = 17$, $s_y = 27$, $r = -0.06$.

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Famous data examples

Aspect ratio

Notice that the scatterplots of the historical examples were made using equal aspect ratio, that is the plot space is a SQUARE not a rectangle.

Most software and most scatterplots you see use unequal aspect ratio. This is an ERROR that is ubiquitous!

When showing two variables, to read association effectively REQUIRES the SQUARE (1:1) aspect ratio.

These slides have more than 50 uses of `aspect.ratio=1`.

Make sure you don't fall into the trap of failing to control aspect ratio.

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Did you notice these interesting patterns?

- Some sports have no data for height, weight
- The positive association between height and weight is visible across sports
- Nonlinear in wrestling?
- An outlier in judo, and football, and archery
- Maybe flatter among swimmers

09:53

- Taller in basketball, volleyball and handball
- Shorter in athletics, weightlifting and wrestling
- Little variance in tennis players

Refine plots to make comparisons easier:

- Remove sports with missings
- Overlay model to assess linear relationships
- Drill down into other strata, eg male/female athletes
- Compare in small chunks, one group against the rest

Note: Because the focus is now on males vs females association shape within sport, make plots scale separately.

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Activity: Olympics

💡 Your turn

- Load the individual athlete data for the Summer 2012 Olympic Games (downloaded from the Guardian's **London 2012 Olympics data**)

```
1 data(oly12, package = "VGAMdata")
```
- Make a scatterplot of **Height** vs **Weight**. You'll see a warning about missing values being removed – what features do you notice? Look for a linear relationship, outliers, discretization, and overplotting (there are more than 10,000 athletes)
- Are there sport(s) where you would expect a different relationship between height and weight?
- Count how many athletes compete in each **Sport**. Are the categories well organised, or would you group some of them together?

Code

Learn

▶ Code

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- Athletics category should have been broken into several more categories like track, field: a shot-putter has a very different physique to a sprinter.

Common ways to augment scatterplots (1/2)

Modification	Example	Purpose
Generally, clustering of male/female athletes		
Overlay model		focus on the trend
model + data		trend plus variation
density		overall distribution, variation and clustering

alpha-blend tall light female weightlifter, tall light male volleyball, divers, cyclists are tiny.

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Modification	Example	Purpose
filled density		high density locations in distribution (modes), variation and clustering
colour		relationship with conditioning and lurking variables

modes and labelling outliers
beyond the largest region.

- Code
- Code
- Code

Common ways to augment scatterplots (2/2)



R

Hexagon binning

Colour maps to the count of points falling in each hexagonal bin, alleviating overplotting.

Bubble plot

Size of each point maps to an extra (weight) variable.

High density regions

The `gghdr` package colours points by highest density regions (HDRs), revealing

Activity: Comparing association and spread

Your turn

1. Using `oly12`, filter to female athletes in Swimming, Archery, Basketball, Handball, Hockey, Tennis, Weightlifting, and Wrestling. Fit a linear trend of `Weight` on `Height` for each `Sport` and plot them together. How does the *association* between height and weight differ across sports?
2. Now filter to female athletes in Shooting, Modern Pentathlon, and Basketball, and overlay density contours of `Height` and `Weight` by `Sport`. How does the *spread* of the relationship differ across sports?
3. Some of this association may be due to unmeasured (lurking) variables. For example, “Athletics” combines throwing and running events, which have quite different body types. If you were only given `Height` and `Weight`, without `Sport`, could you have detected this hidden structure? What pattern might reveal it?
4. Make a scatterplot and overlaid density contours (try a couple of different bandwidths) of `Height` vs `Weight` for `Sport == "Athletics"` only. Can you detect

any multimodality that would hint at the hidden sport subgroup structure?

[Code](#) [Learn](#)

► Code

- Weightlifters are much heavier relative to height, swimmers are leaner, and tennis players are mixed (shorter athletes tend to be heavier, taller athletes tend to be lighter): the *association* between height and weight differs by sport.
- Modern pentathlon athletes are uniformly related in height and weight, while shooters are quite varied in body type: the *spread* of the relationship also differs by sport.
- Some of the association is due to unmeasured variables. For example, “Athletics” masks different body types in throwing vs running events — this is a **lurking** variable.

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- With only **Height** and **Weight**, the hidden structure is hard to detect directly, but it may appear as **multimodality** in the joint distribution — as seen (faintly) in the density contours for Athletics. It's not easy to detect the presence of the additional variable, and thus accurately describe the relationship between height and weight among Olympic athletes.

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Correlation

- Correlation between variables x_1 and x_2 , with n observations in each.

$$r = \frac{\sum_{i=1}^n (x_{i1} - \bar{x}_1)(x_{i2} - \bar{x}_2)}{\sqrt{\sum_{i=1}^n (x_{i1} - \bar{x}_1)^2 \sum_{i=1}^n (x_{i2} - \bar{x}_2)^2}} = \frac{\text{covariance}(x_1, x_2)}{(n-1)s_{x_1}s_{x_2}}$$

- Test for statistical significance, whether population correlation could be 0 based on observed r , using a t_{n-2} distribution:

$$t = \frac{r}{\sqrt{1-r^2}} \sqrt{n-2}$$

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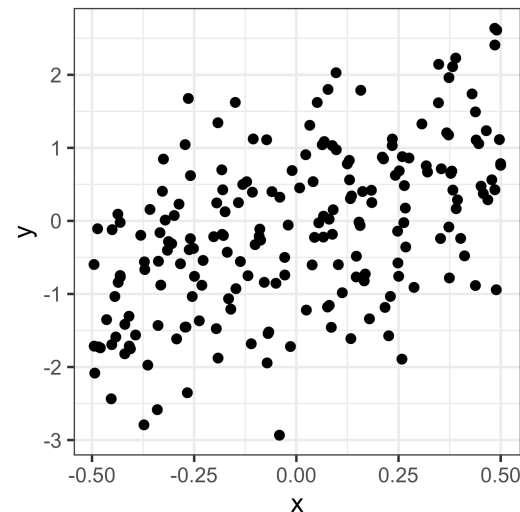
Numerical measures of association

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► Code

[1] 0.52

► Code

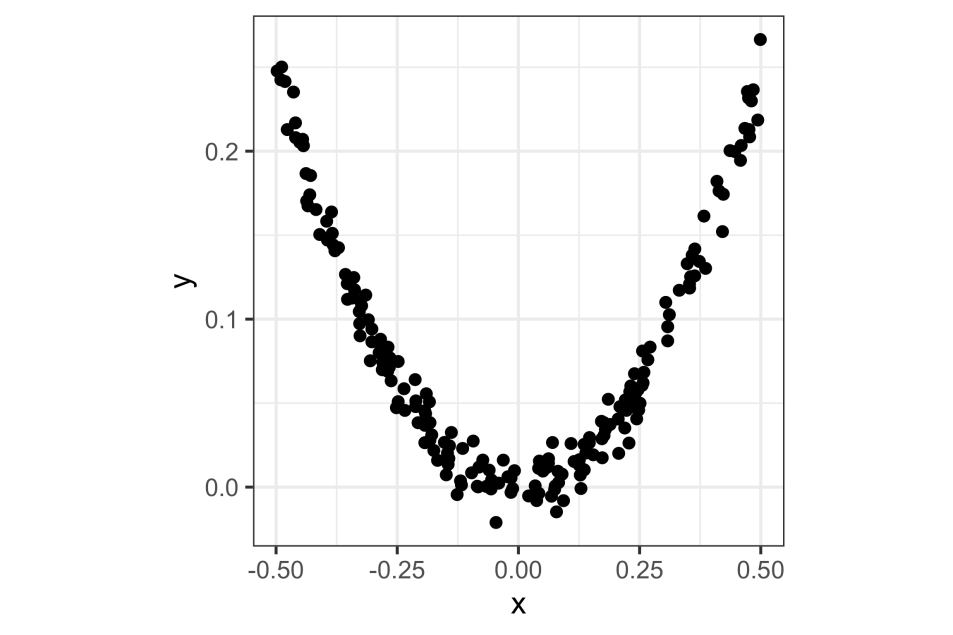
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Problems with correlation (1/2)

Pearson's product-moment correlation

```
data: d1$x and d1$y
t = 9, df = 198, p-value = 2e-15
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 0.41 0.62
sample estimates:
 cor
0.52
```



► Code

```
[1] -0.05
```

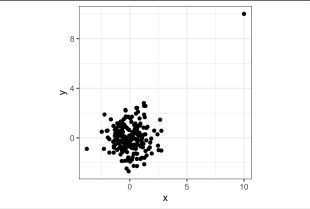
► Code

Pearson's product-moment correlation

```
data: d2$x and d2$y
t = -0.7, df = 198, p-value = 0.5
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
-0.187 0.089
sample estimates:
 cor
-0.05
```

It does not summarise non-linear associations.

Problems with correlation (2/2)



All observations

```
$estimate
cor
0.3

$statistic
t
4.4

$p.value
[1] 1.6e-05
```

Without outlier

```
$estimate
cor
-0.012
```

```
$statistic
t
-0.17

$p.value
[1] 0.87
```

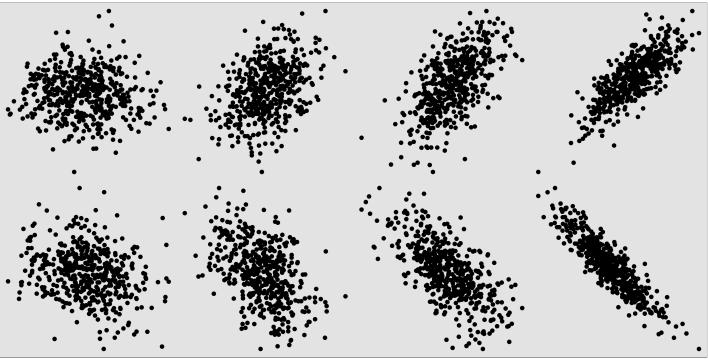
It is affected by extreme values.

Perceiving correlation



[answers](#) [R](#)

Let's play a game: Guess the correlation!



$r = 0.0$	$r = 0.4$	$r = 0.6$	$r = 0.8$
$r = -0.2$	$r = -0.5$	$r = -0.7$	$r = -0.9$

Generally, people don't do very well at this task. Typically people under-estimate r from scatterplots, particularly when it is around 0.4-0.7. The variation in a scatterplot perceptually doesn't vary is not linearly with r . When someone says *correlation is 0.5 it sounds impressive*. BUT when someone shows you a *scatterplot of data that has correlation 0.5*, you will say that's a *weak relationship*.

► Code

Robust correlation measures (1/2)

- Spearman (based on ranks)
- Sort each variable, and return rank (of actual value)
- Compute correlation between ranks of each variable

► Code

```
# A tibble: 7 × 4
  x     y   xr   yr
<dbl> <dbl> <dbl> <dbl>
1  0.7 -1.7    5    2
2  0.5  1.1    4    6
3 -0.6  0.3    2    4
4 -0.2 -0.9    3    3
5 -1.7  0.4    1    5
6  3    -3    6    1
7 10    10    7    7
```

► Code

```
[1] 0.79
```

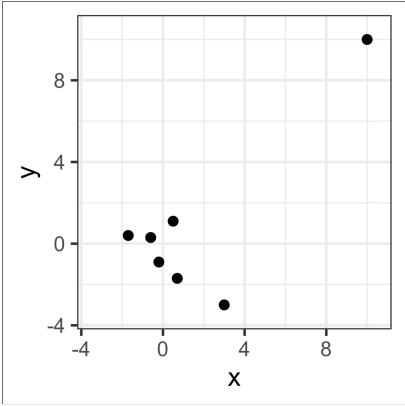
► Code

```
[1] -0.036
```

► Code

```
[1] -0.036
```

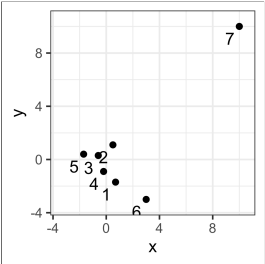
► Code



Robust correlation measures (2/2)

- Kendall τ (based on comparing pairs of observations)
- Sort each variable, and return rank (of actual value)
- For all pairs of observations $(x_i, y_i), (x_j, y_j)$ compare ranks to determine if **concordant**, relatively $x_i < x_j, y_i < y_j$ or $x_i > x_j, y_i > y_j$, or **discordant**, $x_i < x_j, y_i > y_j$ or $x_i > x_j, y_i < y_j$.

$$\tau = \frac{n_c - n_d}{\frac{1}{2}n(n - 1)}$$



```
# A tibble: 7 × 4
  x     y   xr   yr
<dbl> <dbl> <dbl> <dbl>
1  0.7 -1.7    5    2
2  0.5  1.1    4    6
3 -0.6  0.3    2    4
4 -0.2 -0.9    3    3
5 -1.7  0.4    1    5
6  3    -3    6    1
7 10    10    7    7
```

► Code

```
[1] 0.79
```

► Code

[1] -0.14

Robust calculation corrects outlier problems, but nothing measures the non-linear association.

Comparison of correlation measures

sample	corr	spearman	kendall
	0.52	0.512	0.355
	-0.05	-0.087	-0.073
	0.30	-0.023	-0.014

Transformations

for skewness, heteroskedasticity and linearising relationships, and to emphasize association

Circle of transformations for linearising

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Remember the power ladder:

-1, 0, 1/3, 1/2, 1, 2, 3, 4

- 1. Look at the shape of the relationship.
- 2. Imagine this to be a number plane, and depending on which quadrant the shape falls in, you either transform x or y , up or down the ladder: $+, +$ both up; $+, -$ x up, y down; $-, -$ both down; $-, +$ x down, y up

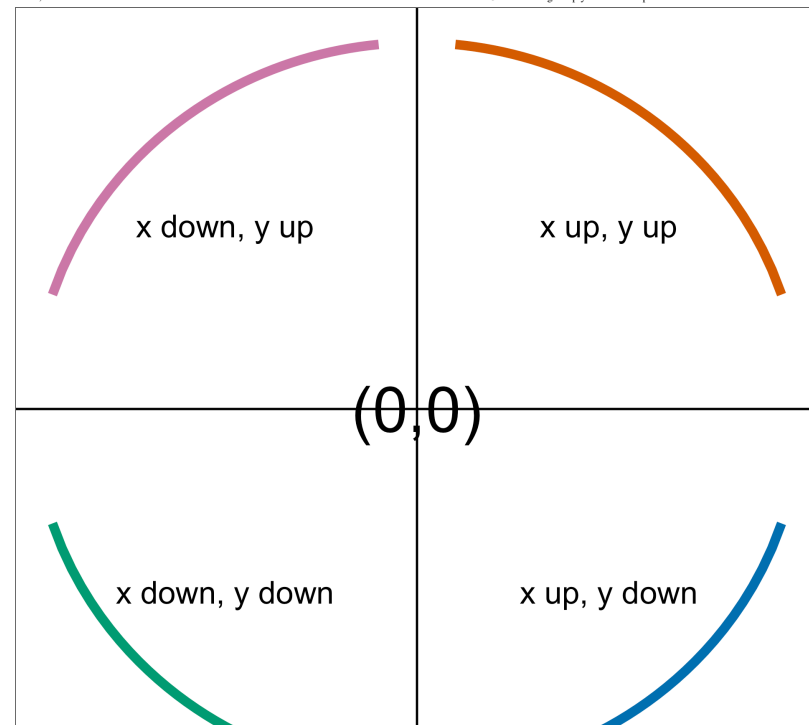
If there is heteroskedasticity, try transforming y , may or may not help

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Activity: Fisherman's Reach crabs

Your turn

1. Load the **Fisherman's Reach crabs data**, collected by measuring 62 mud crabs.
2. Make a scatterplot of **Wgt** by **Length.NSW**. Describe the relationship. (Which variable should be the dependent variable, on the y axis?)
3. Fit a linear model of **Wgt** on **Length.NSW** and plot the residuals. Is there evidence of nonlinearity or heteroskedasticity?
4. Using the circle of transformations, try transforming **Wgt** (e.g. cube root, square root, $\log_{10}$) to linearise the relationship. Which works best, and why?

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5. Is there possibly a lurking variable, that can explain the nonlinearity?


[Code](#)
[Learn](#)

► Code

- Weight is the dependent variable: it should be on the y axis, because weight is related to the volume of the crab, which scales with length^3 , not the reverse.
- The raw scatterplot is positive, strong, and a little nonlinear. Fitting a linear model and examining the residuals reveals a U-shape and clear heteroskedasticity, confirming the nonlinearity.
- Cube root, square root, and $\log_{10}$ transformations of **Wgt** all reduce the curvature to varying degrees, consistent with a physical model where weight scales roughly with length-cubed.

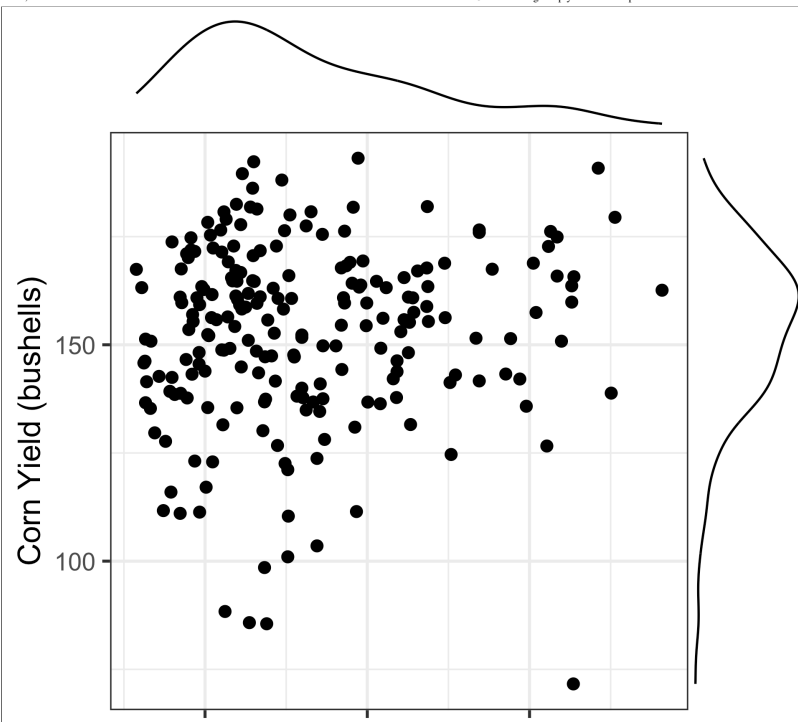
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- Colouring by **Sex** reveals a **lurking variable**: males and females follow somewhat different Length-Weight relationships, and restricting to males and using a cube root transformation gives the best fitting, most linear relationship — matching the physical intuition that weight is proportional to length^3 .

► Code



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3 6 9
Boron (ppm)

[Explanation](#)
[Transform](#)

Interplay between skewness and association

Data is from a soil chemical analysis of a farm field in Iowa. Is there a relationship between Yield and Boron?

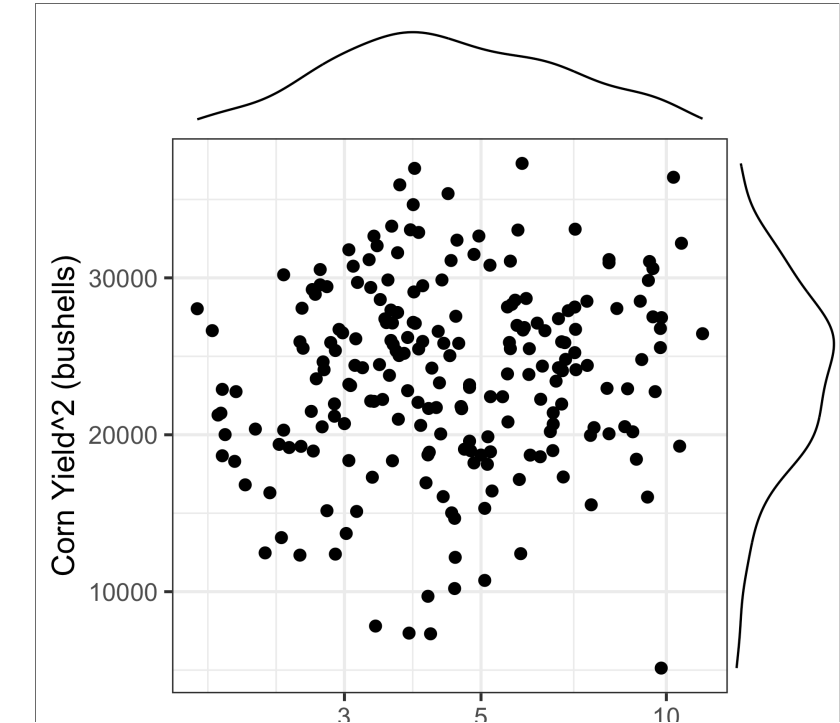
You can get a marginal plot of each variable added to the scatterplot using [ggMarginal](#). This is useful for assessing the skewness in each variable.

Boron is right-skewed Yield is left-skewed. With skewed distributions in marginal variables it is **hard** to assess the relationship between the two. Make a transformation to fix, first.

► Code

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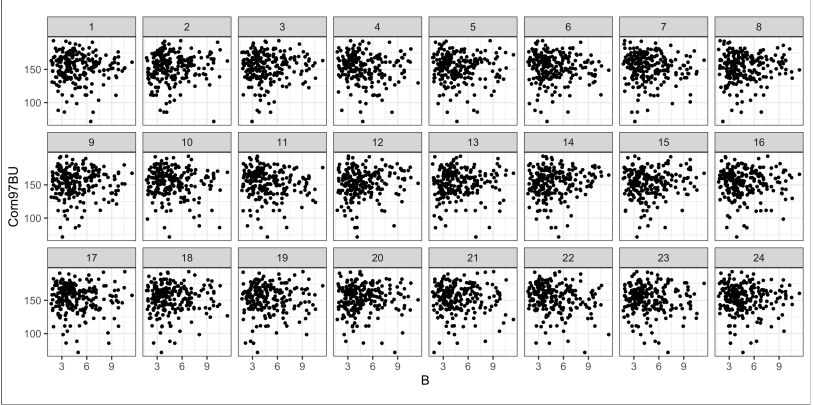
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log Boron (ppm)

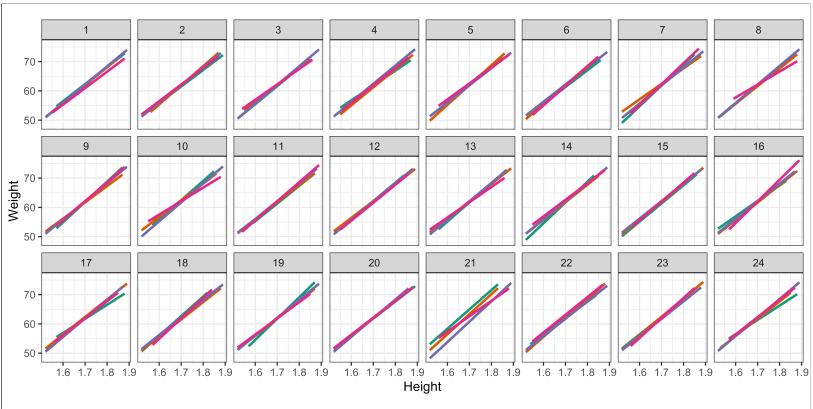
Checking it with visual inference lineup

Soils R Olympics R



► Code

All but one of the panels have had the **association broken** by **permuting** one variable. **There is no association** in these data sets, and hence plots. Does the data plot stand out as being different from the null (no association) plots?



► Code

All but one of the panels have had the **association broken** by **permuting** the Sport label. **There is no difference in the association** between weight and height across

sports in these data sets, and hence plots. Does the data plot stand out as being different from the null (no association difference between sports) plots?

levels are low, there is a positive relationship between yield and iron, with higher iron leading to higher yields.

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Activity: Case study: Soils

Your turn

- 1. Using the baker soils data, make a scatterplot (with marginal densities, and a 2D density overlay) of Corn Yield² (Corn97BU²) against Iron (Fe). Is there a relationship? Could there be a lurking variable?
- 2. Colour the points by whether Calcium (Ca) is high (> 5200 ppm) or low. How does this change your interpretation of the relationship between Iron and Yield?

Code Learn

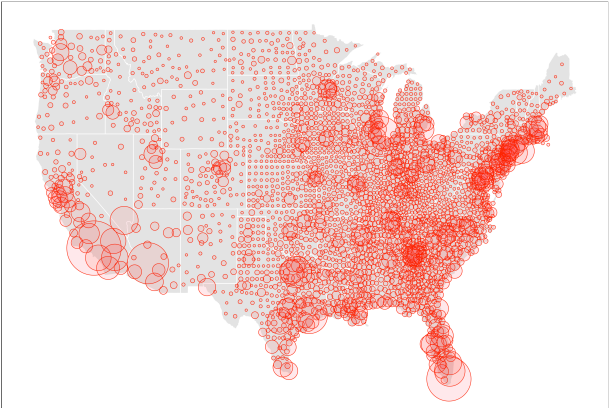
Code

- The raw scatterplot of Yield² against Iron appears to have two different patterns — this hints at a lurking variable.
- Colouring by high vs low Calcium (> 5200 ppm) reveals the structure: if calcium levels in the soil are high, yield is consistently high regardless of iron. If calcium

Scales, benchmarks, calibration matter



about R Learn



This bubble plot here shows total count of COVID-19 incidence (as of Aug 30, 2020) for every county in the USA, inspired by the [New York Times coverage](#).

► Code

Where has COVID-19 hit the hardest?

Where are there more people?

This plot tells you NOTHING except where the population centres are in the USA.

To understand [relative incidence/risk](#), report COVID numbers relative the population.

For example, [number of cases per 100,000 people](#).

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When variables are not quantitative

What do you do if the variables are not continuous/quantitative?

[Type of variable](#) determines the appropriate [mapping](#).

- [Continuous and categorical](#): side-by-side boxplots, side-by-side density plots
- [Both categorical](#): faceted bar charts, stacked bar charts, [mosaic plots](#), [double decker plots](#)

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Beyond quantitative variables

Mosaic plots ^(1/2)

Explain	Plot	Data
-------------------------	----------------------	----------------------

- Built from a [contingency table](#) of two (or more) categorical variables.
- The plot area is recursively split into tiles:
 - Split [horizontally](#) by the first variable, with widths proportional to its marginal counts.
 - Split each column [vertically](#) by the second variable, with heights proportional to the conditional counts.
- Tile [area](#) $\propto$ cell count $\Rightarrow$ area encodes joint frequency.

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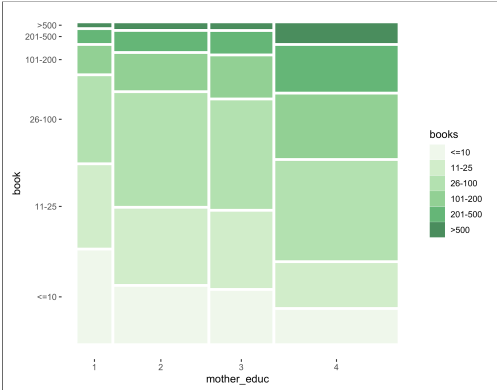
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- Column widths show mother’s education level 4 (finished upper secondary) is the most common in this sample; level 1 is rare.
- Within each column, the split shows the **conditional** distribution of books-at-home category, given mother’s education.
- Students whose mothers have more education more often report **>100** books at home, but with only 47 students, tiles are **unshaded** (grey) — the residuals from independence aren’t large enough to be notable at this sample size.

Read off the proportions of one variable relative to the other.
Number of **books** at home by mother’s education (1 = less than

- ISCED 1, ..., 4 = ISCED 3A), for 2022 PISA students in Turkey (learningtower package).
- ISCED 1: Elementary/primary school completed
 - ISCED 2: Middle school / junior high completed
 - ISCED 3A: Academic high school diploma
- Code



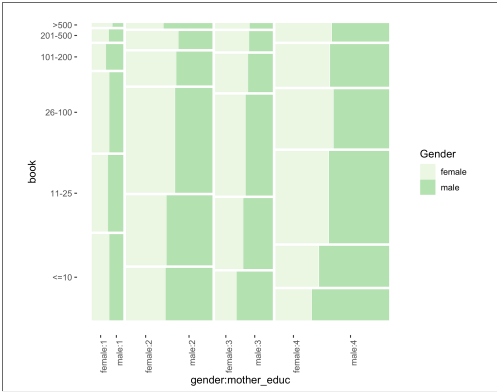
Mosaic plots (2/2)

The third variable would be interleaved on the horizontal axis.
A fourth variable would be interleaved on the vertical axis.
But it gets increasingly difficult to digest the different proportions.

3

4

► Code



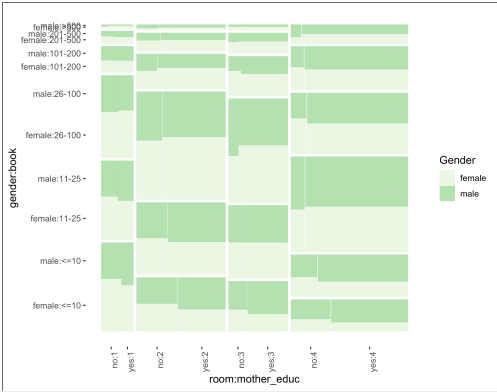
► Code

Double decker plots

Explain **Plot** **Data**

- A **double decker plot** is a mosaic plot where one variable is treated as the **response**, plotted last.
- Explanatory (predictor) categorical variables are combined and shown side by side, with widths proportional to their combined counts.
- Within each combination of explanatory variables, a single bar is split (and coloured) by the response, showing the **conditional probability** of each response category.

Use when: you have a couple of categorical **predictors** and one



categorical **response**, and want to compare how the response's conditional distribution changes across combinations of predictors.

- Column **width** encodes how many students are in that gender / mother's-education combination; relatively few students had mothers who had not finished Year 12.
- The **response** is heavily **imbalanced**: almost all students report having their own room, so most of each bar is orange regardless of group.
- With only 45 students and just 3 without their own room, the small grey slivers (e.g. 1 male student with a mother who didn't finish Year 12) reflect a handful of individuals — not

a reliable estimate of any group difference.

Compared to a mosaic plot, a double decker plot makes it easier to focus on **one response variable** at a time, at the cost of not showing area for the response split as clearly.

Whether a student has their **own room** (grey = no) by **gender** and mother's education, for 2022 PISA students in Australia.

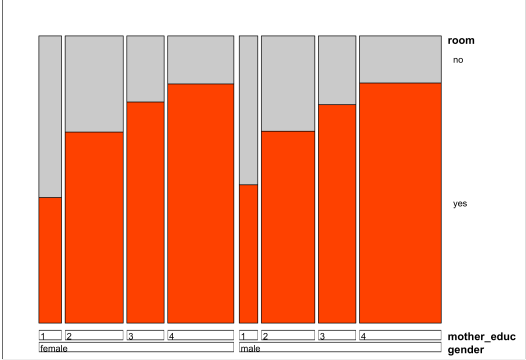
3

3 reorder

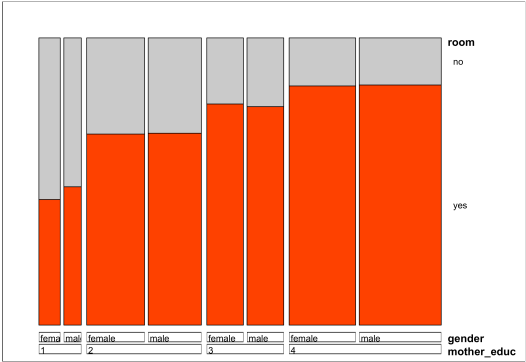
4

4 reorder

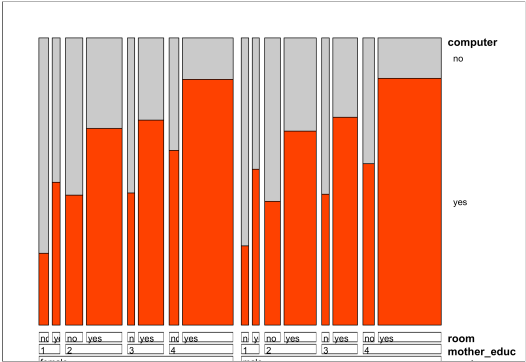
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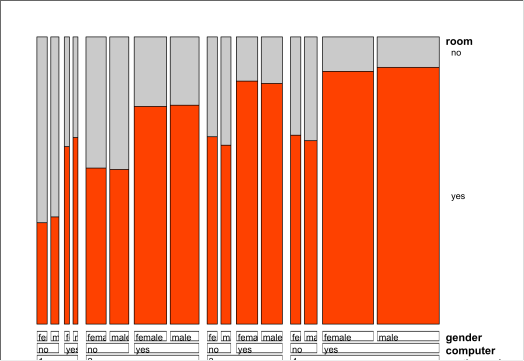
► Code



► Code



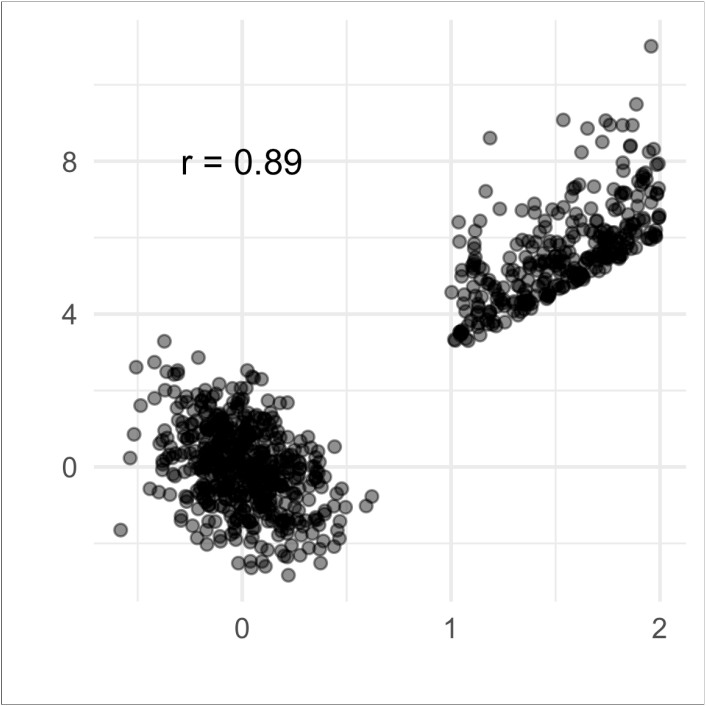
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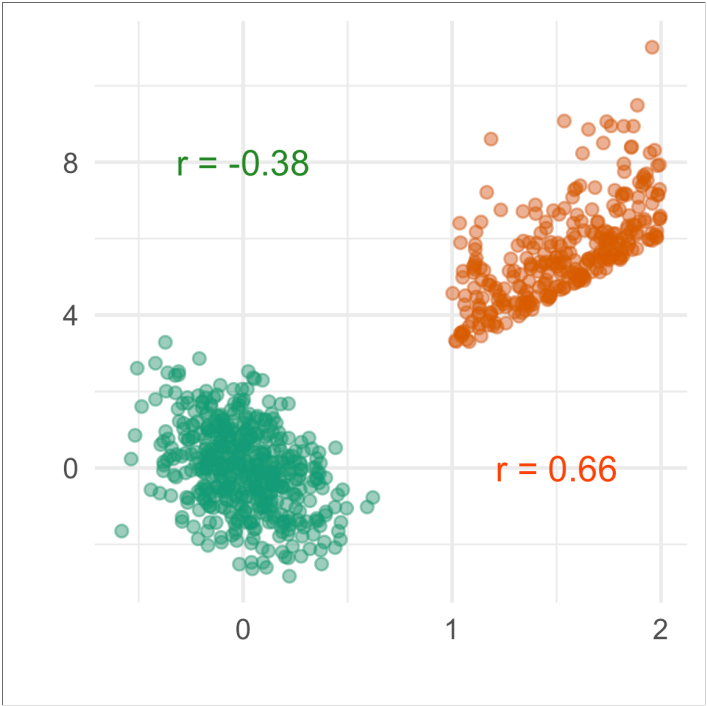


Paradoxes

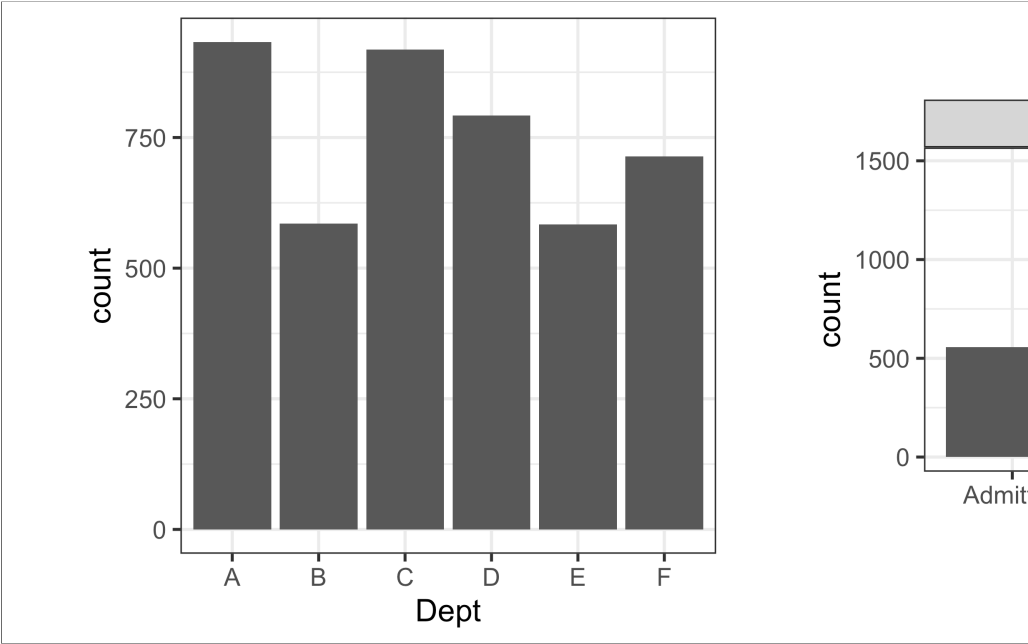
Simpsons paradox

There is an additional variable, which if used for conditioning, changes the association between the variables, you have a **paradox**.





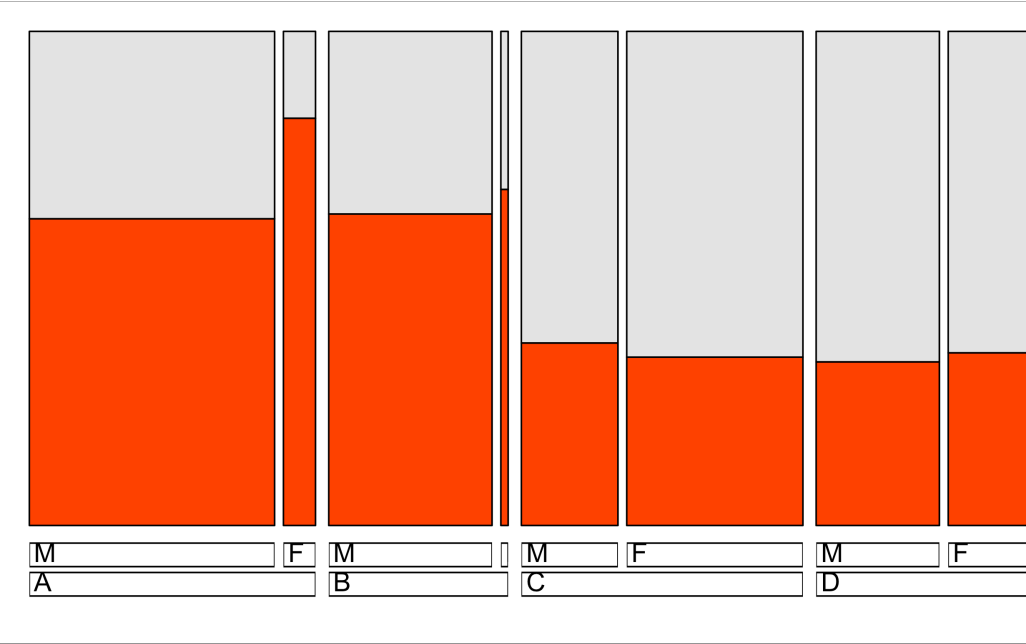
Simpson's paradox: famous example ^(1/2)



Did Berkeley **discriminate** against female applicants?

Example from Unwin (2015)

Simpson’s paradox: famous example ^(2/2)



Based on separately examining each department, there is **no evidence of discrimination** against female applicants.

Example from Unwin (2015)

Always examine the associations in each strata

Handling and imputing missings (1/2)

Check if missings on one variable are related to distribution of the other variable.

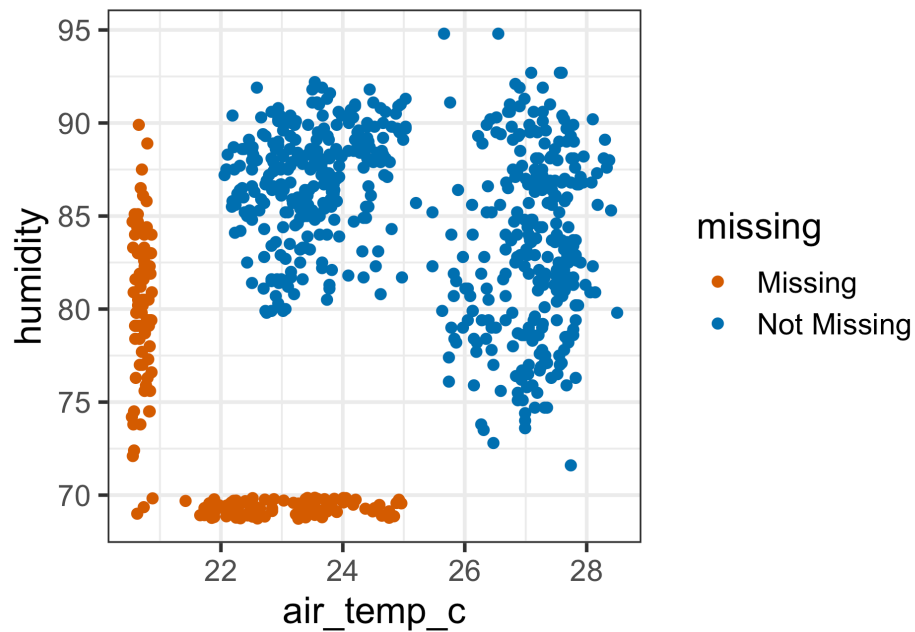
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29/08/2026, 14:32

ETC5521: Diving Deeply into Data Exploration



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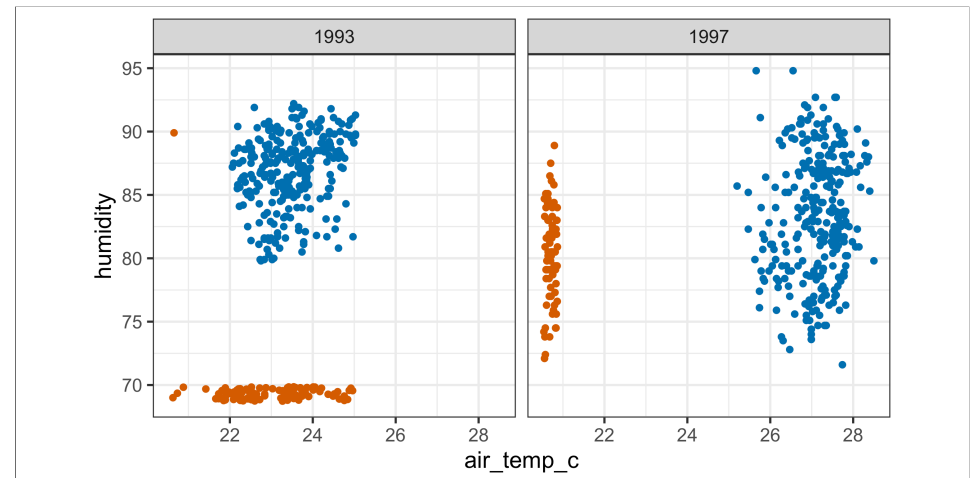
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- Missings plotted in the margins.
- Missings on humidity only occur for lower values of air temperature.
Imputing missings, at least for humidity requires using air temperature values.
But the clustering is due to year

► Code



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Handling and imputing missings (2/2)

Use the mean of complete cases to impute the missings

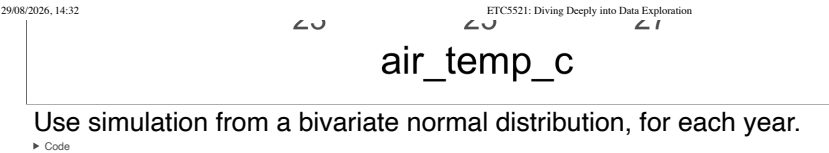
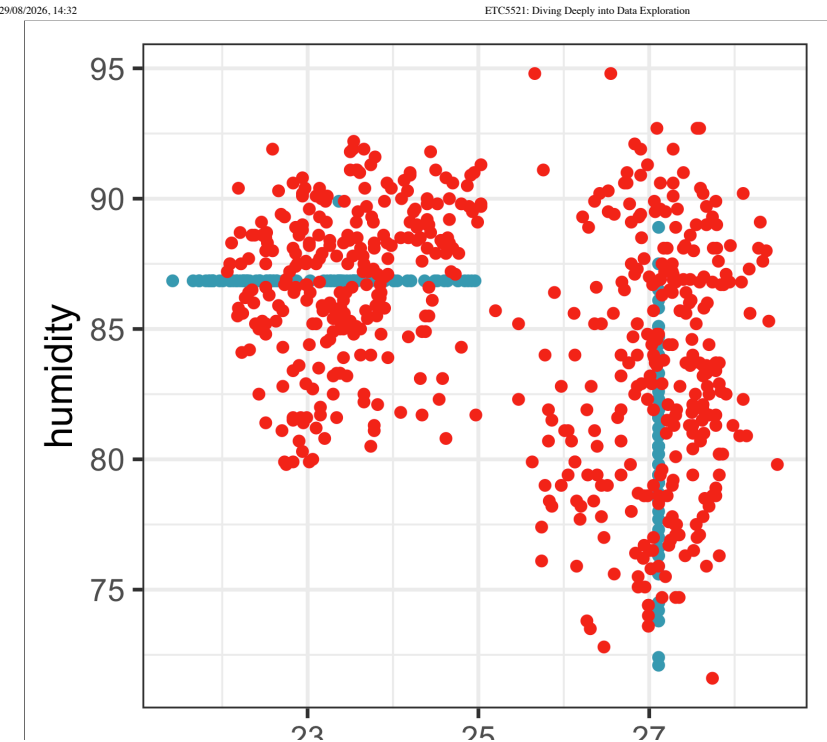
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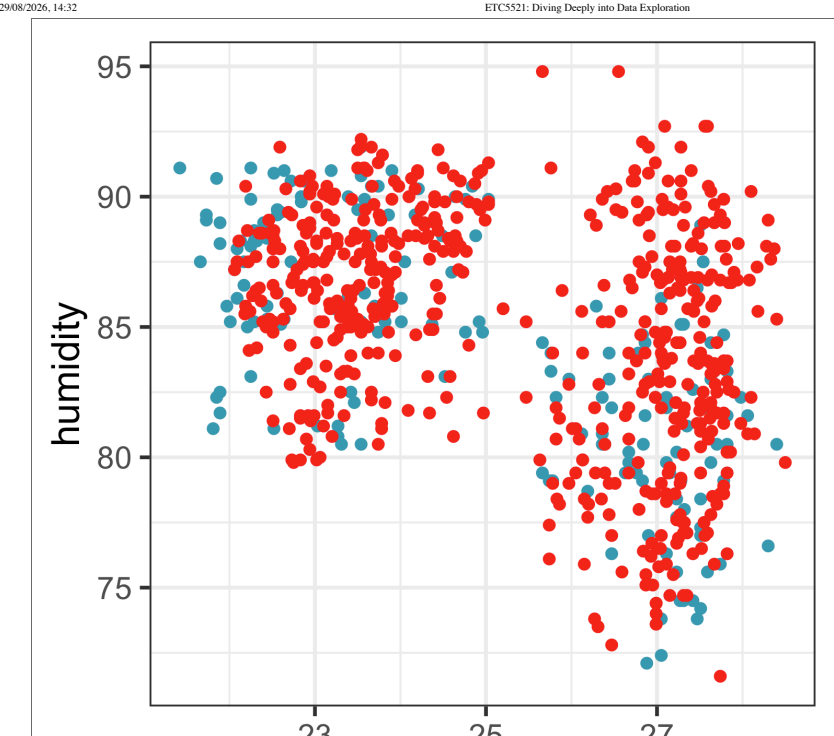
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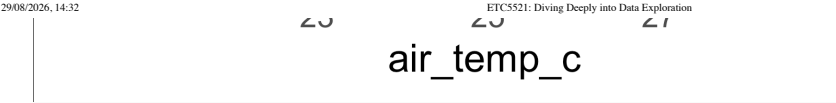
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Networks

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29/08/2026, 14:32 ETC5521: Diving Deeply into Data Exploration

Definition

What do you do if you only receive the association information?
For example, in the Harry Potter series of books, some characters show support and comfort to other characters at different times. (This example is from Sam Tyner's geomnet which was originally taken from The Siena Program.)

► Code

```
Rows: 64
Columns: 4
$ name      <chr> "Adrian Pucey", "Alicia Spinnet", "Ange...
$ schoolyear <int> 1989, 1989, 1989, 1991, 1991, 1989, 198...
$ gender    <fct> M, F, F, M, M, M, M, F, M, M, M, F, M, ...
$ house     <fct> Slytherin, Gryffindor, Gryffindor, Rave...
```

► Code

```
Rows: 434
Columns: 3
$ name1 <chr> "Dean Thomas", "Dean Thomas", "Dean Thomas",...
$ name2 <chr> "Harry James Potter", "Hermione Granger", "N...
$ book  <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,...
```

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Any data where association between multiple pairs of variables can computed can be examined using networks, and explored using exploratory network analysis.

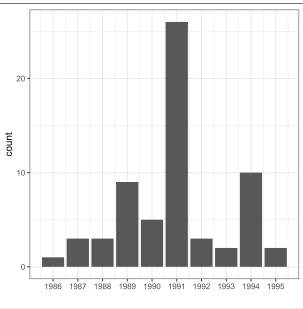
Step 1: explore the node data

school year

house

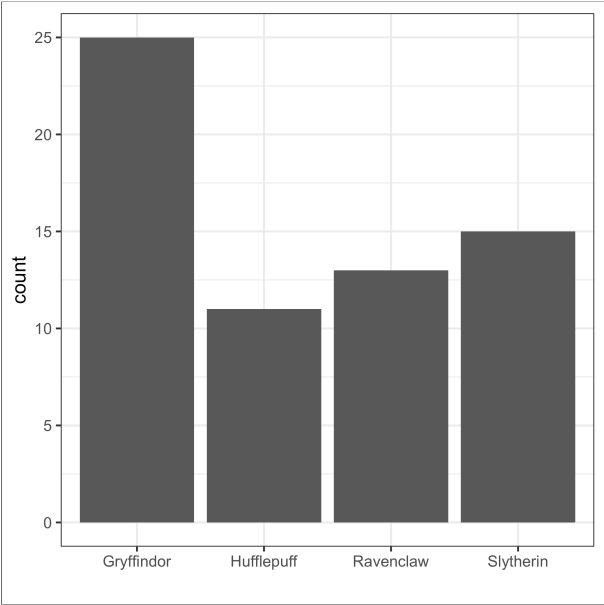
gender

► Code



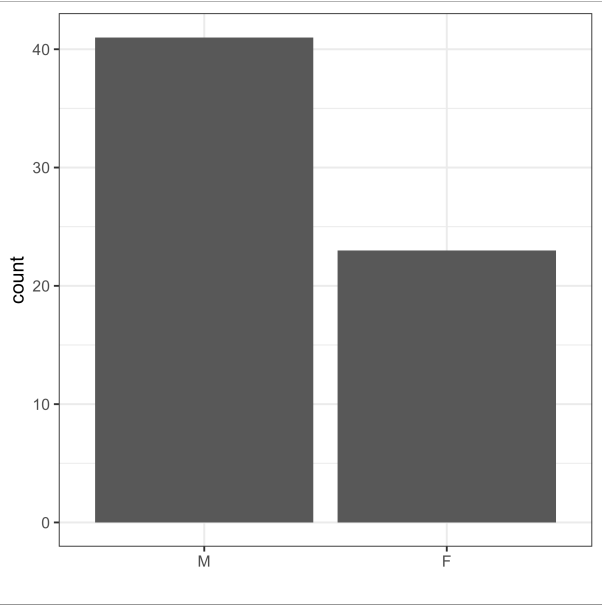
What is school year in HP??

► Code



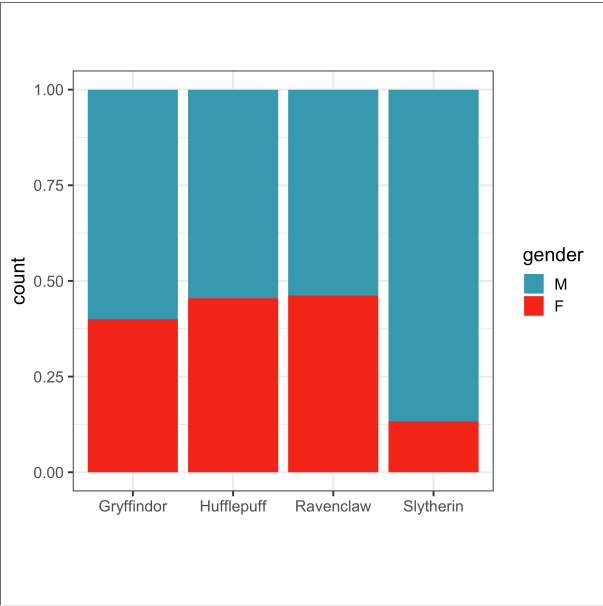
Most of the characters are associated with Gryffindor!!

► Code



Is the book lacking in diversity?

► Code

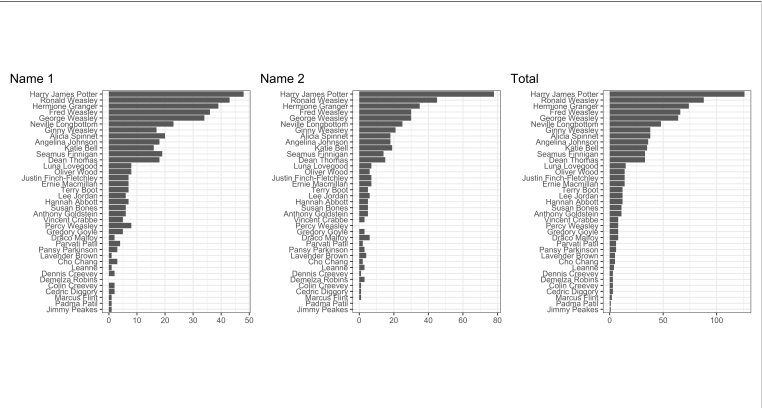


Just Slytherin is lacking diversity.

who is receiving support. Small differences.

Step 2: explore the edge data

► Code



Similar distribution for who is providing support and

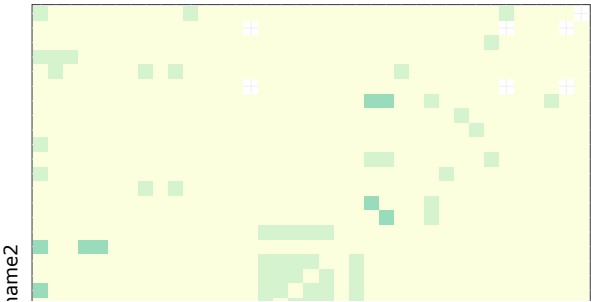
Step 3: Examine association

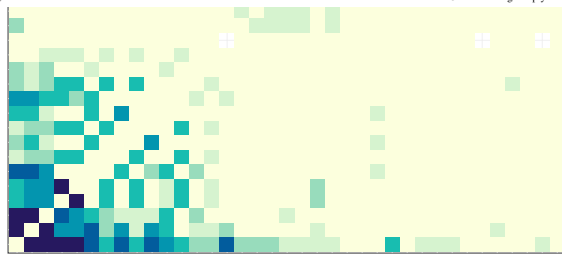
Process the data Heatmap

- Typically only get the connected nodes, so you need to expand the data to include all possible connections, along with the association measure (here 1 or 0).
- Rows/columns ordered by [name](#). Heatmaps need to have rows and columns ordered in a sensible way.

► Code

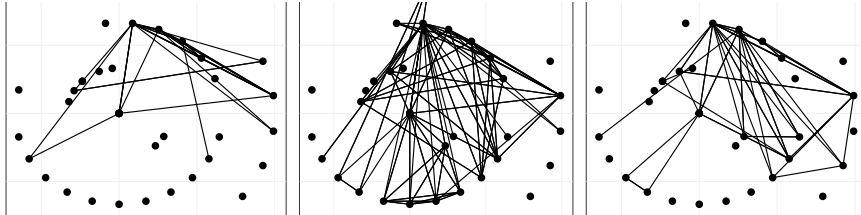
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name1

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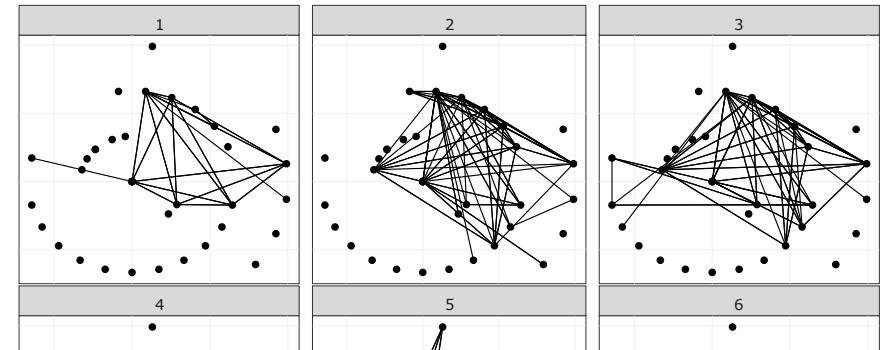
Step 4: Make network summaries

[Process the data](#)
[Network](#)

- Lay out the nodes based on connections over all books
- Set up the data to allow facetting by book, that shows different connections in different books

► Code

► Code



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Key points from today?

- Scatterplots reveal form, strength, and outliers in the relationship between two continuous variables
- Correlation is a single number summary — always look at the plot too, and consider robust measures when the relationship is non-linear or has outliers
- Transformations can linearise a relationship and simplify its description
- Mosaic and double decker plots extend these ideas to categorical variables
- Watch for Simpson's paradox: check associations within subgroups, not just overall
- Missing values need careful handling — understand *why* they're missing before imputing
- Networks let us examine association between entities connected by relationships

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Resources

- Unwin (2015) [Graphical Data Analysis with R](#)
- Graphics using **ggplot2**
- Wilke (2019) [Fundamentals of Data Visualization](#)
- Friendly and Denis [Milestones in History of Thematic Cartography, Statistical Graphics and Data Visualisation](#)
- Tierney et al (2023) [Expanding Tidy Data Principles to Facilitate Missing Data Exploration, Visualization and Assessment of Imputations.](#)
- Sam Tyner's exploring network data with **geomnet**



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Speaker notes